

HW2

Total Points: 50

1. Let $u \in \mathcal{C}((0, \infty) \times \mathbb{R}^d) \cap \mathcal{C}^\infty([0, \infty) \times \mathbb{R}^d)$ be a classical solution to the heat equation, with initial condition satisfying

$$\int_{\mathbb{R}^d} u_0(x) dx = 1.$$

In this problem you may assume that all derivatives of u decays rapidly:

$$\sup_{x \in \mathbb{R}^d} |x|^m |\partial_t^\alpha \partial_{x_1}^{\beta_1} \cdots \partial_{x_d}^{\beta_d} u(t, x)| \leq C_{m, \alpha, \beta, t}, \quad \forall m, \alpha, \beta, t.$$

so differentiation under the integral sign and all integration by parts can be justified.

Let

$$M_k(t) = \int_{\mathbb{R}^d} |x|^k u(t, x) dx, \quad k \geq 0.$$

- (a) [5 points] By differentiating $M_0(t)$ in t , show that $M_0(t) \equiv 1$.
- (b) [5 points] Show that $M_2(t) = M_2(0) + 2dt$.

Remark: this can be interpreted as the root-mean-square displacement of the diffusion particle is $O(\sqrt{t})$.

2. [10 points] Let $u \in \mathcal{C}_c(\mathbb{R}^d)$ with $u \geq 0$ and $u \not\equiv 0$. Let $\rho \in L^1(\mathbb{R}^d) \cap \mathcal{C}(\mathbb{R}^d)$ with $\rho > 0$. Show that

$$(\rho * u)(x) > 0, \quad \forall x \in \mathbb{R}^d.$$

Remark: in particular, $\rho = G_t$ satisfies the hypothesis, and this means that heat equation has infinite speed of propagation.

3. [5 points] Let $f(x) = \mathbb{1}_{[-a, a]}(x)$ where $a > 0$. Compute $\hat{f}(\xi)$.
4. (a) [2 points] Let $f(x) = e^{-a|x|}$, $x \in \mathbb{R}$. Compute $\hat{f}(\xi)$.

Next we consider $f(x) = e^{-a|x|}$, $x \in \mathbb{R}^3$. Here, $|x| := \sqrt{x_1^2 + x_2^2 + x_3^2}$ is the Euclidean norm.

- (b) [3 points] Use the radial symmetry of f to show that $\hat{f}(\xi)$ is also radially symmetric, that is, a function of $|\xi|$.
- (c) [10 points] Let $\xi = (0, 0, \rho)$ where $\rho = |\xi|$. Using spherical coordinates, show that

$$\hat{f}(\xi) = 2\pi \int_0^\infty r^2 e^{-ar} \int_0^\pi e^{-2\pi i \rho r \cos \theta} \sin \theta d\theta dr.$$

Evaluate the two integrals and conclude that

$$\hat{f}(\xi) = \frac{8\pi a}{(a^2 + 4\pi^2|\xi|^2)^2}.$$

5. In this problem you can use the Young's inequality

$$\|f * g\|_{L^r} \leq \|f\|_{L^p} \|g\|_{L^q}, \quad \frac{1}{r} + 1 = \frac{1}{p} + \frac{1}{q}, \quad p, q, r \in [1, \infty].$$

Recall the 1D heat kernel $G_t(x) = \frac{1}{\sqrt{4\pi t}} e^{-\frac{x^2}{4t}}$. Let $u(t, x) = (G_t * u_0)(x)$.

(a) [5 points] Show that

$$\|G_t\|_{L^p} = C_p t^{-\frac{1}{2}(1-1/p)}, \quad p \in [1, \infty].$$

(b) [5 points] For $1 \leq p \leq q \leq \infty$, find the exponent $\beta \geq 0$ such that

$$\|u(t)\|_{L^q} \leq C t^{-\beta} \|u_0\|_{L^p}.$$

For which p, q is $\beta > 0$?